Fall-degree bounds: E12b (ℓ against n at m = 3), relaunch r2. Run report

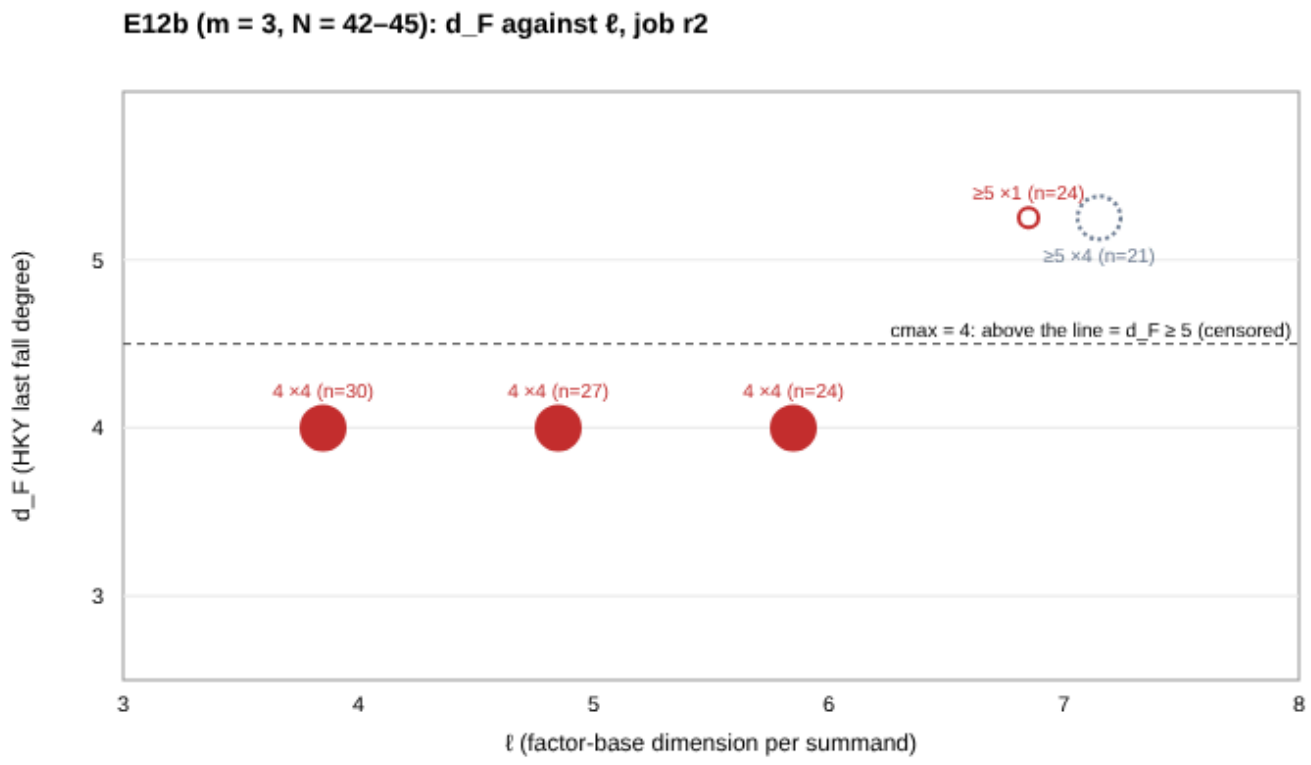
Outcome: inconclusive. Three of four cells completed. The fourth (3:24:7) was cut off by a Spot reclaim after one draw. The one relaunch Amendment 11 allows is now used, so no registered reading is recorded.

What ran. Amendment 11 (E12b) of `research/fall_degree_bounds_20261008/PREREGISTRATION.md`, registered before any E12 draw. Engine `examples/hky_par.rs` at source commit `371035216942`, built on the instance from the S3 archive `fd-src-371035216942.tgz` (sha256 `4ae808ac…`). Launch card `e12/launch3.card.yaml` (receipt `e12/launch_receipt3/`). Job `20261027-181036-fd-e12b-ell-20261027-r2-409f97`: ephemeral Spot m7i.4xlarge (16 vCPU, 61 GB), us-east-1c, lab account. Draw rule: `ladder_sample` with seed 20261012, the first 4 draws with no solutions, `cmax = 4`. No isolation level was earned, so wall times are indications only and are not scored.

Gates, all passed on this instance's binary. 0: the three counter tests pass. 1: `e7/validate.sh` passes. 2–3: `newcells` reproduces `e6/B_5-10-2.rerun`, `e6/B_3-18-6`, `e8/E8_4-8-2` and `e8/E8_5-8-2` (empty diffs). Lean: `lake build` of `FallDegree` passed (`out/lean/status`: `exit=0`).

Interruption. The command exited 143 at 19:06:09 UTC while running 3:24:7 draw 1. The runner logged “interrupted (spot reclaim)”. r1 (`e12/run-b1`) had stopped at the same point.

Graph



Red: E12b r2 (3:24:7 not completed: 1 of 4 draws). Grey dashed: E10 3:21:7 (N = 42, diagonal), for context. Marker area ∝ draws.

Recorded d_F of the 13 r2 draws (red) against ℓ, with the four committed E10 draws of 3:21:7 (ℓ = 7, N = 42, on the diagonal) in grey for context. A hollow “≥5” marker is a draw whose degree-4 closure stopped without reaching 1. Source: `plot_e12b.py`.

Measured values (per draw)

cell m:n:ℓ	N	status	d_F per draw	closure dim / columns at degrees 2, 3, 4	s per draw (16 threads)
3:30:4	42	completed	4, 4, 4, 4	71/904; 3,197/12,384; 124,314/124,314	184–234
3:27:5	42	completed	4, 4, 4, 4	68/904; 2,912/12,384; 124,314/124,314	160–167
3:24:6	42	completed	4, 4, 4, 4	65/904; 2,628/12,384; 124,314/124,314	131–137
3:24:7	45	**not completed (1 of 4)**	**≥5**	68/1,036; 2,842/15,226; 157,926/164,221	298

Reproduction. r2 equals r1 on every non-timing field for all 13 draws: d_F, per-degree dimensions and rounds, `v_basis`, `x_r` and solution counts. Raw records: `e12/run-b2/out/E_*.jsonl`, with gate, build and Lean logs and `host.txt` beside them.

Against the registered rules (Amendment 11)

- **“n-driven”** (3:30:4 and 3:27:5 both have median ≥ 5): **excluded**. Both medians are 4.
- **“Consistent with H_ℓ”** (the $\ell \le 6$ cells ≤ 4, and 3:24:7 at median ≥ 5): the first part holds. 3:24:7 is not completed, and its one finished draw is censored at ≥ 5. The amendment does not say whether one draw makes a median, so the choice between this reading and “mixed” is **left open**. Only a registered amendment or a completed 3:24:7 cell decides it.

Unregistered observation, not a reading. At fixed N = 42, the degree-4 closure reaches the full space (124,314 columns) at $\ell = 4$, 5 and 6. The single $\ell = 7$ draw (N = 45) stops 6,295 columns short, as the diagonal cell 3:21:7 does. Toy size only; no exponent and no security claim.